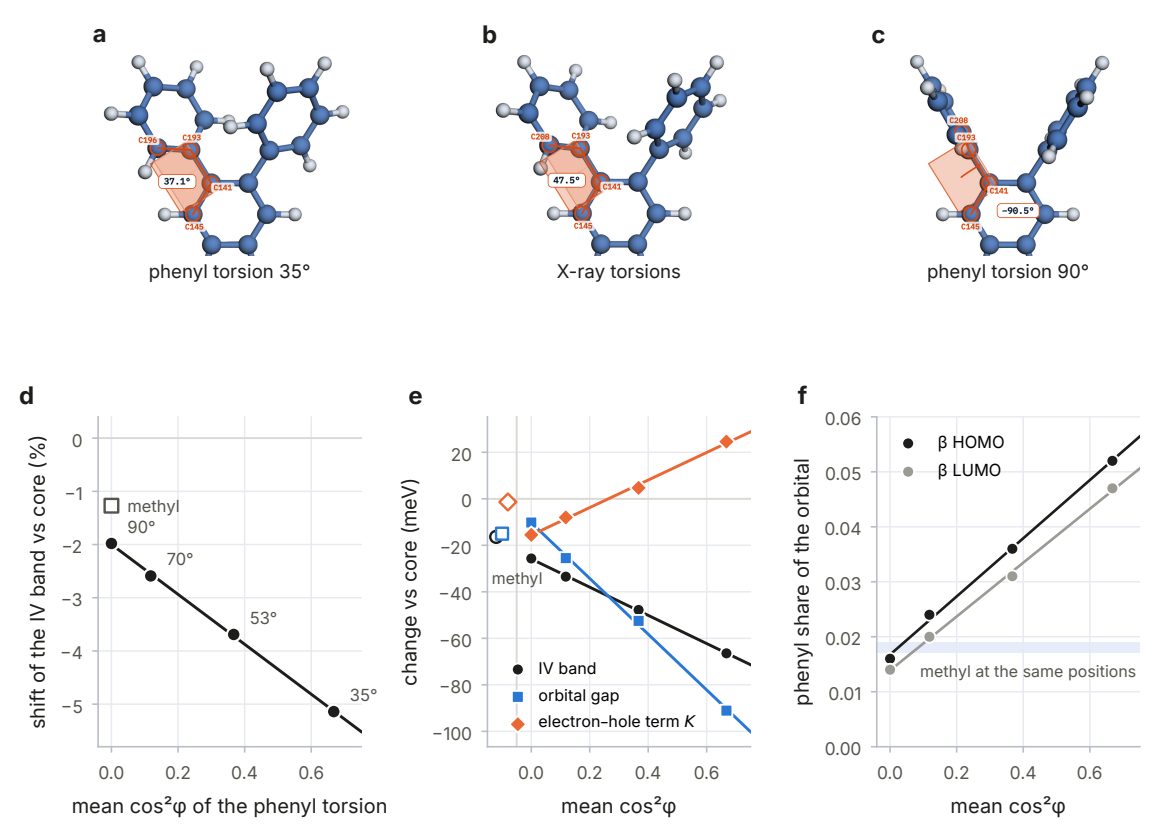

a
C196
C193
37.1°
C141
C145
phenyl torsion 35°
b
C208
C193
47.5°
C141
C145
X-ray torsions
c
C208
C193
C141
−90.5°
C145
phenyl torsion 90°
d
shift of the IV band vs core (%)
0
−1
−2
−3
−4
−5
methyl
90°
70°
53°
35°
0.0
0.2
0.4
0.6
mean cos²φ of the phenyl torsion
e
change vs core (meV)
20
0
−20
−40
−60
−80
−100
methyl
IV band
orbital gap
electron–hole term K
0.0
0.2
0.4
0.6
mean cos²φ
f
phenyl share of the orbital
0.06
0.05
0.04
0.03
0.02
0.01
0.00
β HOMO
β LUMO
methyl at the same positions
0.0
0.2
0.4
0.6
mean cos²φ